Polar Coordinates & Polar Graphs

ANSWER KEY

Section 1: Coordinate Conversions (Polar <-> Rectangular)

1 Convert the polar point $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$ to rectangular coordinates (x, y).

a. $(-3\sqrt{3}, 3)$ ✓

b. $(3\sqrt{3}, -3)$

c. $(-3, 3\sqrt{3})$

d. $(3, -3\sqrt{3})$

$x = r\cos\theta = 6\cos\left(\frac{5\pi}{6}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. $y = r\sin\theta = 6\sin\left(\frac{5\pi}{6}\right) = 6\left(\frac{1}{2}\right) = 3$. The rectangular point is $(-3\sqrt{3}, 3)$.

2 Convert the rectangular point $(-4, 4)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $\left(4\sqrt{2}, \frac{3\pi}{4}\right)$ ✓

b. $\left(4\sqrt{2}, \frac{7\pi}{4}\right)$

c. $\left(8, \frac{3\pi}{4}\right)$

d. $\left(4\sqrt{2}, \frac{5\pi}{4}\right)$

$r = \sqrt{(-4)^2 + 4^2} = \sqrt{32} = 4\sqrt{2}$. Since $(-4, 4)$ is in Quadrant II, $\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$.

3 Which polar point represents the same location as $(r, \theta) = \left(3, \frac{\pi}{3}\right)$?

a. $\left(-3, \frac{4\pi}{3}\right)$ ✓

b. $\left(-3, \frac{\pi}{3}\right)$

c. $\left(3, -\frac{\pi}{3}\right)$

d. $\left(-3, \frac{2\pi}{3}\right)$

In polar coordinates, $(r, \theta) = (-r, \theta + \pi)$. For $\left(3, \frac{\pi}{3}\right)$, $\left(-3, \frac{\pi}{3} + \pi\right) = \left(-3, \frac{4\pi}{3}\right)$ represents the exact same physical point.

Section 2: Converting Equations between Polar and Cartesian

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4 Convert the polar equation $r = 6\cos\theta$ into rectangular coordinates.

a. $(x-3)^2 + y^2 = 9$ ✓

b. $x^2 + (y-3)^2 = 9$

c. $x^2 + y^2 = 6x$

d. $x^2 + y^2 = 36$

Multiply both sides by r: $r^2 = 6r\cos\theta$. Substitute $r^2 = x^2 + y^2$ and $x = r\cos\theta$:
$x^2 + y^2 = 6x \Rightarrow x^2 - 6x + 9 + y^2 = 9 \Rightarrow (x-3)^2 + y^2 = 9$ (circle centered at $(3, 0)$ of radius 3).

5 Convert the rectangular equation $x^2 + y^2 = 16$ to polar form.

a. $r = 4$ ✓

b. $r = 16$

c. $r = 4\cos\theta$

d. $r = 16\sin\theta$

Since $r^2 = x^2 + y^2$, we have $r^2 = 16 \Rightarrow r = 4$.

6 Convert $r = 4\csc\theta$ to rectangular coordinates.

a. $y = 4$ ✓

b. $x = 4$

c. $y = 4x$

d. $x^2 + y^2 = 16$

Multiply both sides by $\sin\theta$: $r\sin\theta = 4$. Since $y = r\sin\theta$, this is the horizontal line $y = 4$.

Section 3: Rose Curves and Symmetry

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7 How many petals does the rose curve $r = 5\cos(4\theta)$ have?

a. 8 ✓

b. 4

c. 16

d. 2

For a rose curve $r = a\cos(n\theta)$, if n is even, there are $2n$ petals. Here $n = 4$ is even, so the rose has $2(4) = 8$ petals.

8 How many petals does the rose curve $r = 3\sin(5\theta)$ have?

a. 5 ✓

b. 10

c. 15

d. 3

If n is odd, the rose curve has exactly n petals. Since $n = 5$ is odd, the curve has 5 petals.

9 What is the maximum distance from the pole (origin) to the curve $r = 4\sin(3\theta)$?

a. 4 ✓

b. 12

c. 3

d. 8

The maximum value of $|\sin(3\theta)|$ is 1. Therefore, the maximum radius is $|r| = 4(1) = 4$.

Section 4: Limaçons and Special Polar Curves

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10 Classify the polar curve $r = 2 + 2\cos\theta$.

a. **Cardioid (heart-shaped limaçon)** ✓

b. Limaçon with an inner loop

c. Dimpled limaçon

d. Convex limaçon

For $r = a \pm b\cos\theta$, when $\frac{a}{b} = 1$, the curve is a cardioid.

11 Which polar equation produces a limaçon with an inner loop?

a. $r = 2 + 4\cos\theta$ ✓

b. $r = 4 + 2\cos\theta$

c. $r = 3 + 3\sin\theta$

d. $r = 5 + 4\cos\theta$

A limaçon $r = a + b\cos\theta$ has an inner loop when $\frac{a}{b} < 1$. For $r = 2 + 4\cos\theta$, $\frac{2}{4} = 0.5 < 1$.

12 At what angles $\theta \in [0, 2\pi)$ does the curve $r = 2 + 4\cos\theta$ pass through the pole ($r = 0$)?

a. $\theta = \frac{2\pi}{3}$ **and** $\theta = \frac{4\pi}{3}$ ✓

b. $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$

c. $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$

d. $\theta = \pi$

Set $r = 0 \Rightarrow 2 + 4\cos\theta = 0 \Rightarrow \cos\theta = -\frac{1}{2}$. In $[0, 2\pi)$, this occurs at $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$.

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